

UNIVERSIDAD POLITÉCNICA DE AGUASCALIENTES

BACHELOR'S DEGREE IN ENGINEERING TECHNOLOGIES IN
INFORMATION TECHNOLOGY AND DIGITAL INNOVATION



Sprint #2 Deliverable 3

KAVEM System - Client Management Module

COURSE

Integrative Project II

Period

April - August 2026

TEAM #3

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Contents

1	Introduction	2
2	Functional and Non-Functional Requirements	3
2.1	Functional Requirements (FR)	3
2.2	Non-Functional Requirements (NFR)	3
2.3	Color Palette	4
3	Use Case Diagram	6
4	Use Case Scenario Cards	8
4.1	UC-01: Register New Client	8
4.2	UC-02: Update Data from an Existing Client	8
4.3	UC-03: Delete Customer Registration	9
4.4	UC-04: General Consultation and Filtered Customer Search	9
5	Database Entity-Relationship Diagram	10
5.1	Main Entities	10
6	Interface Design (Mockups)	11
6.1	Main Client Menu Mockup	11
6.2	Client Details Mockup	11
7	Navigation Design	13
7.1	Navigation Structure	14
8	Test Cases	15
8.1	Test Cases for Client Management	15
8.2	Test Summary for Sprint #2	19
9	Screenshots of the Program in Operation	20
9.1	Screenshot 1: Client List Screen	20
9.2	Screenshot 2: Client Registration	20
9.3	Screenshot 3: Client Details with Order History	21
10	Sprint Management Using Jira	22
10.1	Sprint #2 Overview	22
10.2	Jira Board Screenshot	22
10.3	Tasks Completed in Sprint #2	22
10.4	Task Assignments Summary	23
10.5	Burndown Chart	23

Introduction

The following content presents the results and advances of Sprint #2. In this sprint, we focused on the Client Management module. We worked on the design and development of functionalities such as registering new clients, updating existing client information, deleting client records, and consulting the client list with search filters. The project was managed using Jira.

Functional and Non-Functional Requirements

Functional Requirements (FR)

- **FR - Customer Data Management:** The system must allow the registration and storage of the following mandatory fields for the customer profile: Customer/Company Name, Contact Name, Email Address, Delivery Address, Telephone Number.
- **FR - Input Validations (Integrity Rules):** To mitigate billing and data-entry errors, the system must automatically validate the following constraints in real time before allowing data to be saved:
 - Customer Name: maximum 15 characters, minimum 5 characters, mandatory, not exclusively numeric.
 - Contact Name: maximum 50 characters, minimum 3 characters.
 - Email Address: must contain @ character, maximum 50 characters.
 - Address: maximum 50 characters.
 - Telephone Number: numeric characters only (restrict letters and special symbols).
- **FR - Order History:** The system must link each customer to their respective historical log of quotations and purchases to facilitate tracking and recurring sales.

Non-Functional Requirements (NFR)

- **NFR01 - Corporate Identity (Usability):** The interface dedicated to the customer module must visually incorporate the company logo and the established institutional color palette.
- **NFR02 - Intuitive Interface Design (UX):** To achieve a more familiar and agile workflow for employees, the system must employ recognizable, standardized icons on navigation links and buttons instead of extensive text labels.

Color Palette



Figure 1: KAVEM Color Palette: #E49C54, #E6BE8C, #F2B06E, #F5ECE3

Use Case Diagram

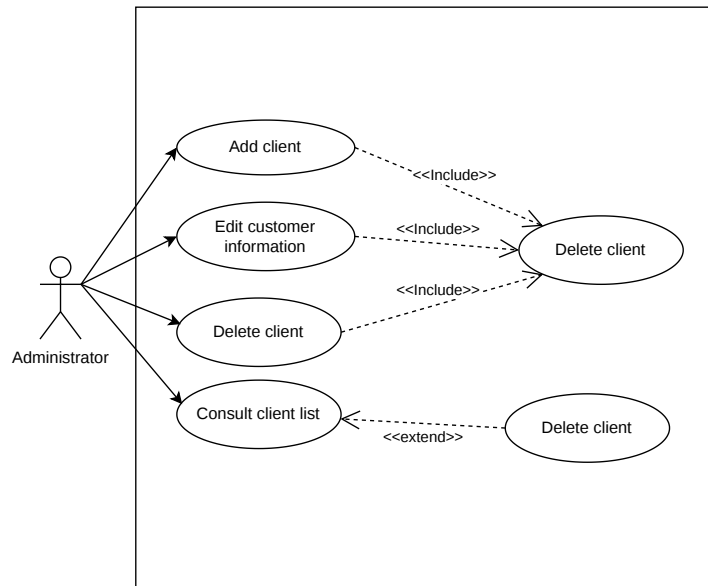


Figure 2: Use Case Diagram for Client Management Module

Main Actors:

- Administrator

Main Use Cases:

- Register New Client
- Update Data from an Existing Client
- Delete Customer Registration
- General Consultation and Filtered Customer Search
- Authenticate User (extends Register, Update, Delete, and Consult)

Use Case Scenario Cards

UC-01: Register New Client

Element	Description
Scenario	The administrator registers a new distributor or recurring customer
Main Actor	Administrator
Precondition	The Administrator must have valid access credentials and the client's commercial data
Main Flow	1. The Administrator selects the "Add Client" option 2. The system requests identity authentication (Authenticate User) 3. After successful validation, the system displays the registration form 4. The Administrator enters the data (Name, RFC, Mail, Telephone, Address) 5. The Administrator confirms by clicking "Save" 6. The system validates the data, stores the record, and displays a success message
Alternate Flow	Authentication Error: If credentials fail, system denies access
Postcondition	The new customer is active in the system and available for assigning orders

UC-02: Update Data from an Existing Client

Element	Description
Scenario	The Administrator modifies a customer's contact or delivery information
Main Actor	Administrator
Precondition	The selected client must exist in the database
Main Flow	1. The Administrator selects a specific client and clicks "Edit Information" 2. The system invokes Authenticate User 3. The system enables editable fields with current customer information 4. The Administrator makes modifications 5. The Administrator presses "Update" 6. The system saves changes and refreshes the customer's profile
Alternate Flow	Empty required fields: system highlights error in red and blocks saving
Postcondition	Customer data is updated in the database immediately

UC-03: Delete Customer Registration

Element	Description
Scenario	The Administrator deregisters a customer from the catalog
Main Actor	Administrator
Precondition	The selected client must exist in the database
Main Flow	1. The Administrator selects the client and presses "Delete" 2. The system requests security validation through Authenticate User 3. The system displays a confirmation pop-up 4. The Administrator confirms the deletion 5. The system performs a logical erase and confirms the action
Alternate Flow	Customer with Active Orders: system prevents deletion and issues notice
Postcondition	The customer no longer appears on active sales and administration lists

UC-04: General Consultation and Filtered Customer Search

Element	Description
Scenario	The Administrator views the complete customer catalog or uses filters to locate a client
Main Actor	Administrator
Main Flow	1. The Administrator clicks "Check Client List" 2. The system requires Authenticate User validation 3. The system renders a table with all registered clients 4. (Optional) Administrator types Name or ID in search bar 5. The system dynamically filters the table in real time 6. The Administrator locates the desired client
Alternate Flow	No matches: system displays "No customers were found"
Postcondition	Information is presented on the screen ready for reading

Database Entity-Relationship Diagram

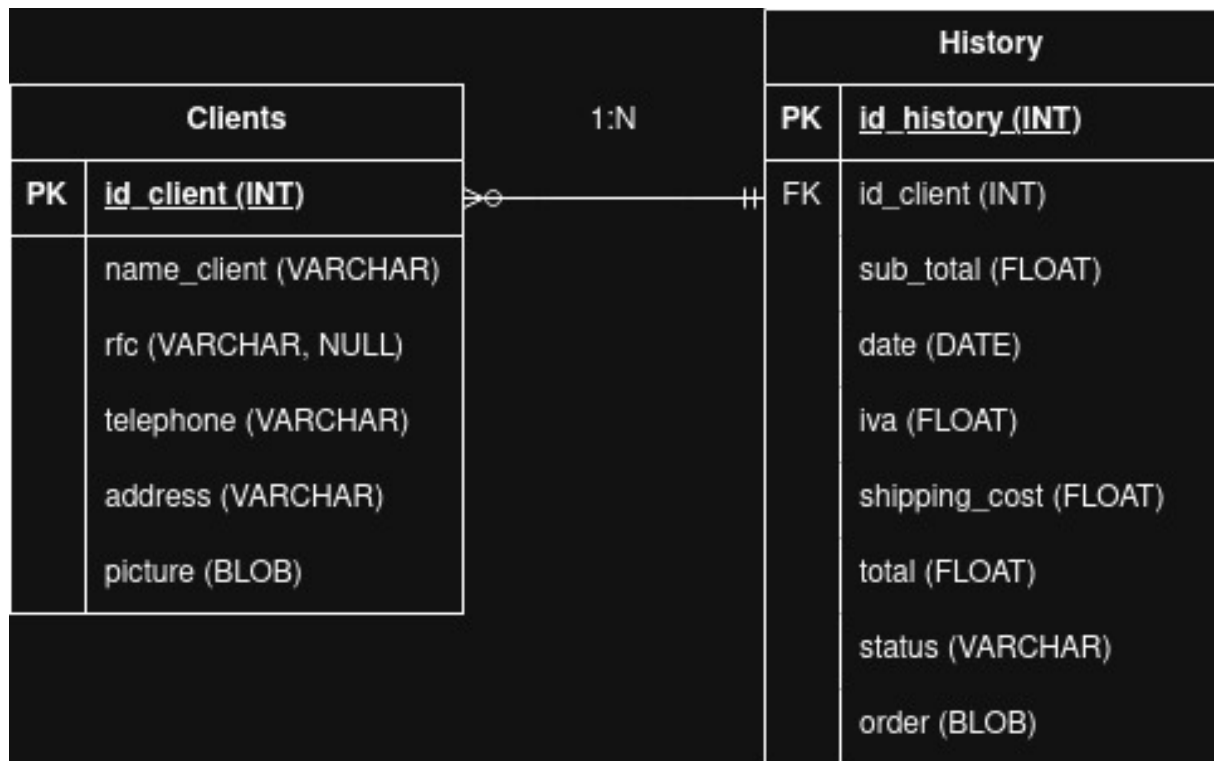


Figure 3: Entity-Relationship Diagram for Client Management Module

Main Entities

- **Clients:** id_client (PK), name_client, rfc, telephone, address, picture
- **History:** id_history (PK), id_client (FK), sub_total, date, iva, shipping_cost, total, status, order

Interface Design (Mockups)

Main Client Menu Mockup



Figure 4: Main Client Menu - Inventory, Clients, History with Filters

Client Details Mockup



Figure 5: Client Details Screen showing Name, Contact, Email, Phone, and Order History

Navigation Design

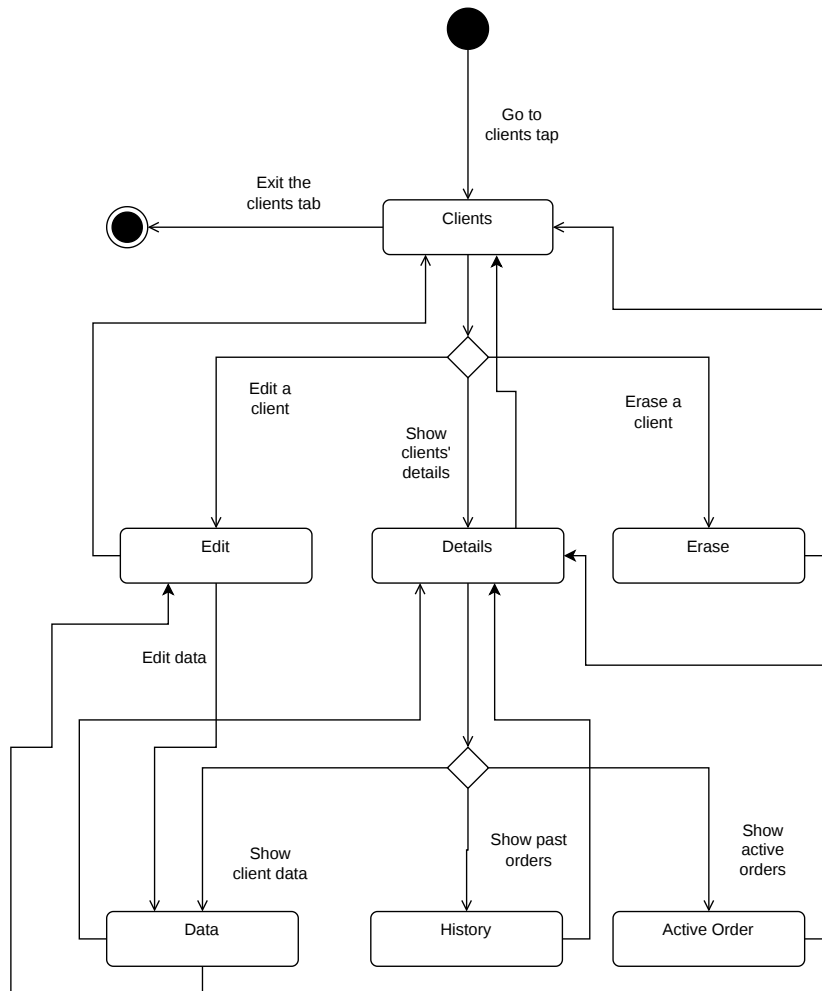


Figure 6: Navigation Flow Diagram for Client Module

Navigation Structure

- **Main Menu**

- Go to Clients tab

- * View Client List

- Apply Filters (Filter 1, 2, 3, 4)
 - Search by Name/ID
 - Select Client
 - View Client Details
 - Edit Client Information
 - Delete Client (with confirmation)
 - View Order History

- * Add New Client

- Fill Registration Form
 - Save Client

Test Cases

Sprint #2 Focus: Client Management operations (Register, Update, Delete, Search)

Test Cases for Client Management

Test Case ID	TC-CLIENT-01
Module	Client Management
Test Title	Verify successful client registration
Priority	High
Preconditions	Administrator is authenticated
Test Data	Name: "John Doe", Contact: "John", Email: "john@company.com", Phone: "4491234567", Address: "Main Street 123"
Execution Steps	1. Click "Add Client" 2. Fill all fields with valid data 3. Click "Save"
Expected Results	System saves data and displays green pop-up notification "Client created successfully"
Actual Results	As expected. A green toast banner reading "Client created successfully" displayed instantly. Entry verified in database.
Status	Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 24, 2026

Test Case ID	TC-CLIENT-02
Module	Client Management
Test Title	Verify successful client update
Priority	High
Preconditions	Client exists in database (ID: 1024)
Test Data	Change name from "John Doe" to "Jane Smith"
Execution Steps	1. Navigate to Client List 2. Search for ID 1024 3. Click "Edit" 4. Modify name to "Jane Smith" 5. Click "Save Changes"
Expected Results	System displays "Changes saved successfully" and redirects to client summary

Actual Results	Re-	As expected. Toast notification triggered and client view updated immediately
Status		Pass
Tester		Alan Ricardo Jauregui De La Riva
Date		May 24, 2026

Test Case ID	TC-CLIENT-03
Module	Client Management
Test Title	Verify phone number validation (letters not allowed)
Priority	High
Preconditions	Editing client ID 1024
Test Data	Phone: "449abc1234"
Execution Steps	1. Open edit modal for client ID 1024 2. Clear phone contents and type "449abc1234" 3. Click "Save Changes"
Expected Results	System rejects submission and shows inline warning "Phone number must not include letters"
Actual Results	FAIL: Application displayed inline error but submission logic was not disabled. Corrupted string was sent to server and saved.
Status	Don't Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 24, 2026

Test Case ID	TC-CLIENT-04
Module	Client Management
Test Title	Verify successful client deletion
Priority	High
Preconditions	Client exists with no active orders (ID: 1025)
Test Data	Client ID: 1025
Execution Steps	1. Navigate to client listing page 2. Locate client ID 1025 3. Select "Delete" option 4. Confirm deletion in modal
Expected Results	Modal confirmation, then "Client deleted successfully" toast banner

Actual Results	Re-	As expected. Deletion popup prompted confirmation; row dropped dynamically with success message
Status		Pass
Tester		Alan Ricardo Jauregui De La Riva
Date		May 24, 2026

Test Case ID	TC-CLIENT-05
Module	Client Management
Test Title	Verify deletion blocked when client has active orders
Priority	High
Preconditions	Client has active order in progress (ID: 1024)
Test Data	Client ID: 1024
Execution Steps	1. Search for client ID 1024 2. Click "Delete" button 3. Observe modal restriction behavior
Expected Results	System blocks delete action and shows warning: "Cannot delete client because they have an active order in progress"
Actual Results	As expected. System threw operational block notification modal detailing order dependency conflict. Record preserved.
Status	Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 24, 2026

Test Case ID	TC-CLIENT-06
Module	Client Management
Test Title	Verify viewing client details
Priority	Medium
Preconditions	Client exists in database
Test Data	Client ID: 1024
Execution Steps	1. Open Client Directory 2. Select client row 1024 3. Tap "View Details" 4. Audit rendered fields
Expected Results	Profile card opens with Name, Phone, Email, Address, and Registration Date

Actual Results	Re-	As expected. Card loads smoothly with all relevant data fields neatly aligned
Status		Pass
Tester		Alan Ricardo Jauregui De La Riva
Date		May 24, 2026

Test Case ID	TC-CLIENT-07
Module	Client Management
Test Title	Verify error when viewing non-existent client
Priority	Medium
Preconditions	None
Test Data	Client ID: 9999 (does not exist)
Execution Steps	1. Force route to '/clients/9999' 2. Monitor viewport response
Expected Results	Application shows error message: "Error: The client does not exist or was deleted"
Actual Results	FAIL: Application gets stuck in infinite loop showing loading spinner. No fallback error handling.
Status	Don't Pass
Tester	Alan Ricardo Jauregui De La Riva
Date	May 24, 2026

Test Case ID	TC-CLIENT-08
Module	Client Management
Test Title	Verify viewing order history
Priority	Medium
Preconditions	Client has historical orders
Test Data	Client ID with order history
Execution Steps	1. Open client details 2. Select "Order History" tab 3. Check historic list parameters
Expected Results	History populates showing past orders with ID, Date, Total, and Status
Actual Results	As expected. Historic ledger loads exactly 5 archived logs matching database rows
Status	Pass
Tester	Alan Ricardo Jauregui De La Riva

Date	May 24, 2026
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Test Summary for Sprint #2

Test Case ID	Status
TC-CLIENT-01	Pass
TC-CLIENT-02	Pass
TC-CLIENT-03	Don't Pass
TC-CLIENT-04	Pass
TC-CLIENT-05	Pass
TC-CLIENT-06	Pass
TC-CLIENT-07	Don't Pass
TC-CLIENT-08	Pass

Table 9: Test Cases completed in Sprint #2

Screenshots of the Program in Operation

Screenshot 1: Client List Screen

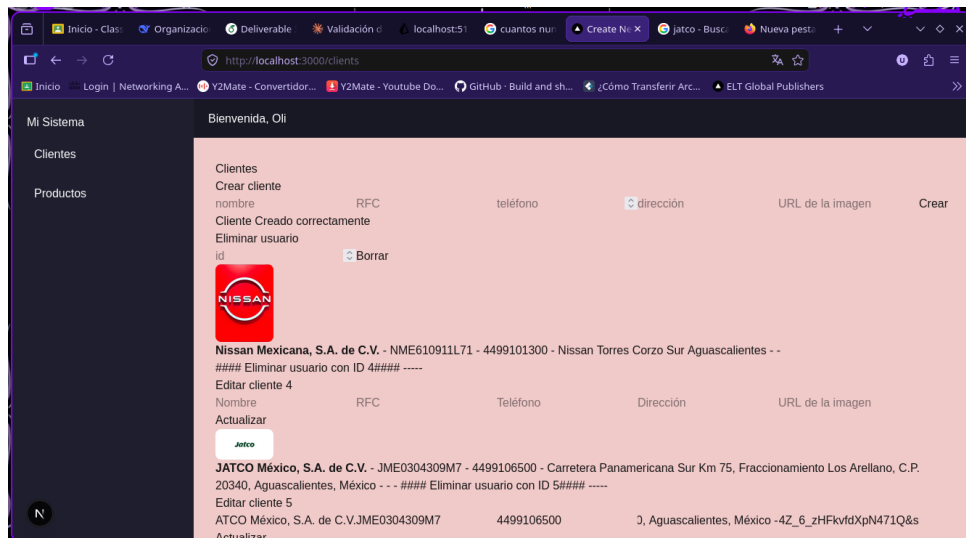


Figure 7: Client list screen showing filters and registered clients

Screenshot 2: Client Registration

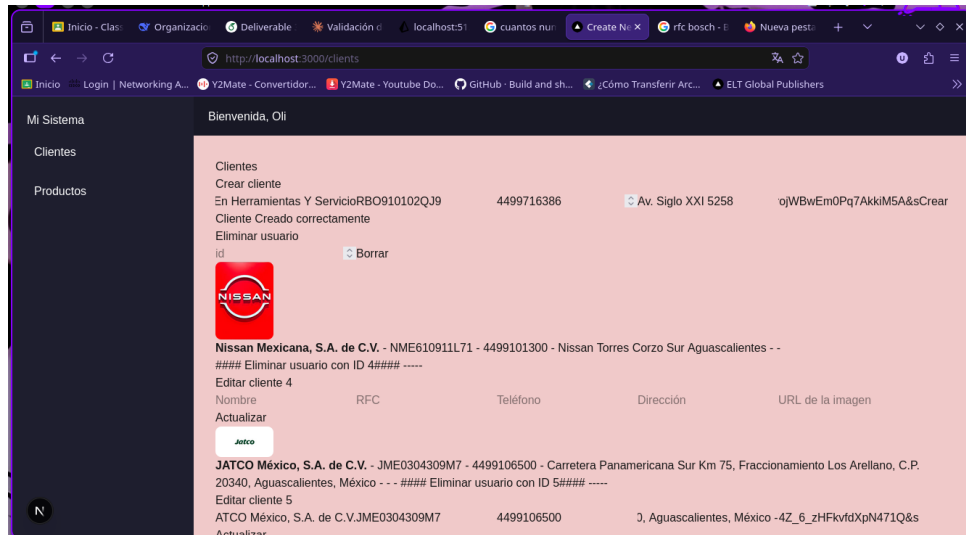


Figure 8: Client registration form

Screenshot 3: Client Details with Order History

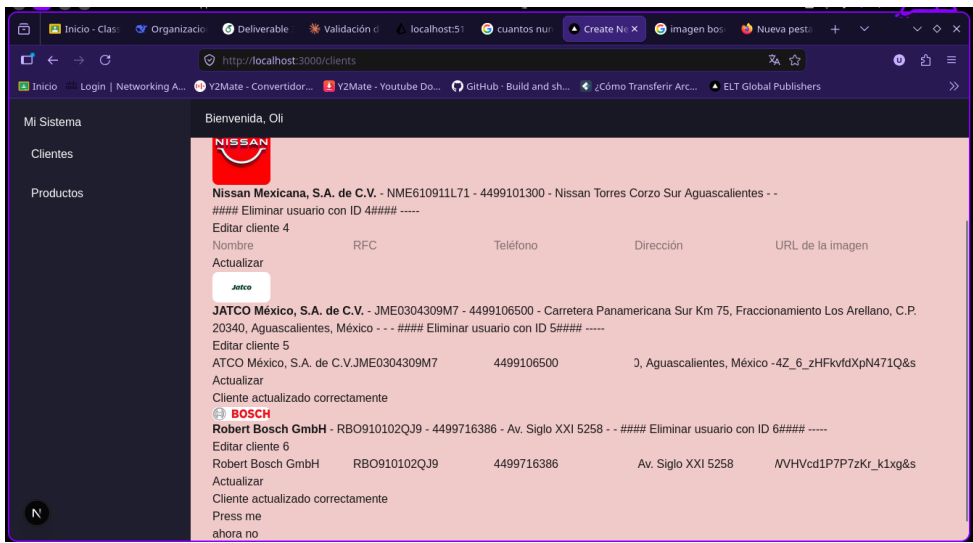


Figure 9: Client details screen showing information and order history

Sprint Management Using Jira

Sprint #2 Overview

Sprint Duration	May 21 – May 24, 2026
Total Tasks	8
Completed Tasks	8
Completion Percentage	100%

Jira Board Screenshot

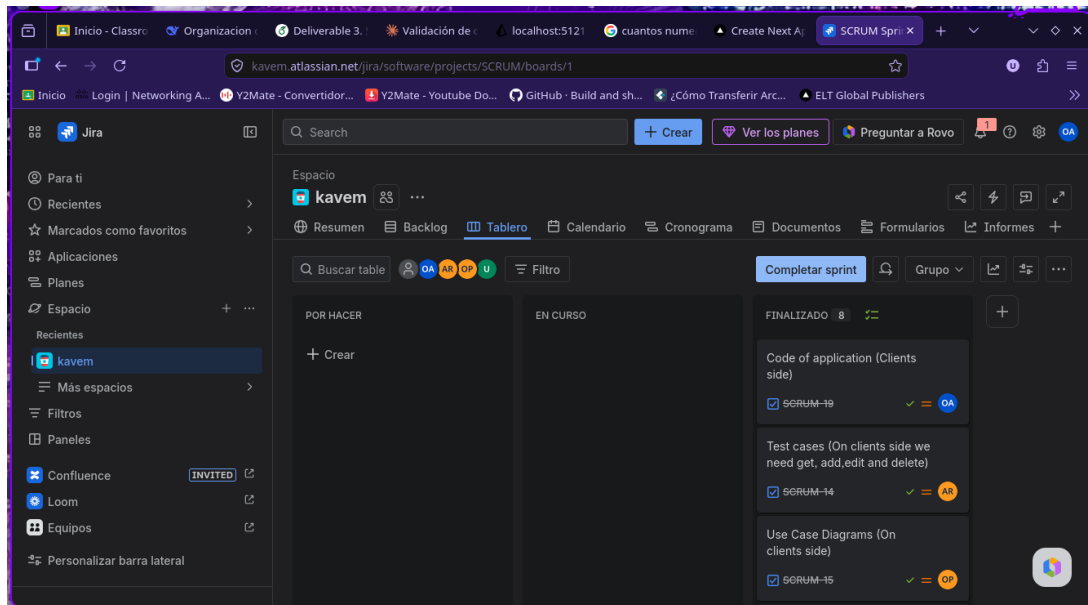


Figure 10: Jira Sprint Board for Sprint #2

Tasks Completed in Sprint #2

Issue ID	Task Name	Assigned To	Status
SCRUM-14	Client Requirements (Functional & Non-Functional)	Noé Ezequiel Estrada Ramírez	Finalizado
SCRUM-15	Client Use Case Diagram	Omar Facio Palos	Finalizado
SCRUM-16	Client Scenario Cards	Omar Facio Palos	Finalizado
SCRUM-17	Client ER Diagram	Noé Ezequiel Estrada Ramírez	Finalizado
SCRUM-18	Client Mockups (UI Design)	Alan Ricardo Jauregui De La Riva	Finalizado
SCRUM-19	Client Navigation Design	Alan Ricardo Jauregui De La Riva	Finalizado

SCRUM-20	Client Test Cases	Alan Ricardo Jauregui De La Riva	Finalizado
SCRUM-21	Client Module Code	Olivia Lizbeth Chairez Alba	Finalizado

Table 10: Tasks completed during Sprint #2

Task Assignments Summary

- **Olivia Lizbeth Chairez Alba (UP240594)** - Client Module Code (SCRUM-21)
- **Noé Ezequiel Estrada Ramírez (UP240770)** - Client Requirements (SCRUM-14), Client ER Diagram (SCRUM-17)
- **Omar Facio Palos (UP240470)** - Client Use Case Diagram (SCRUM-15), Client Scenario Cards (SCRUM-16)
- **Alan Ricardo Jauregui De La Riva (UP240960)** - Client Mockups (SCRUM-18), Client Navigation (SCRUM-19), Client Test Cases (SCRUM-20)

Burndown Chart

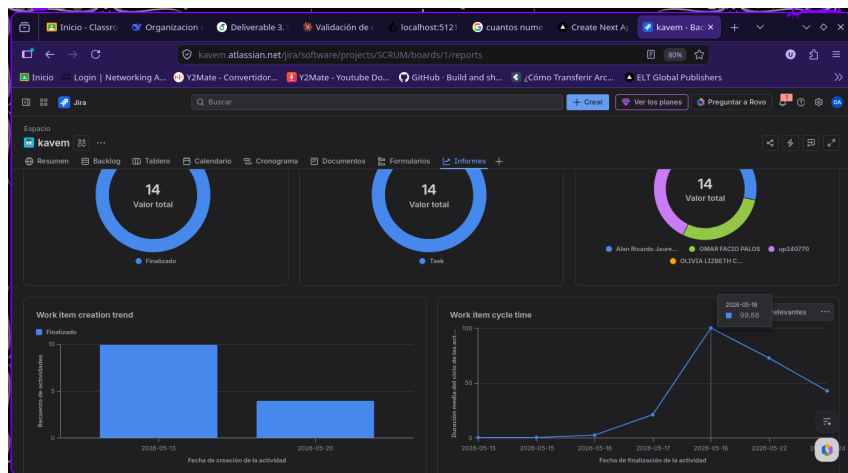


Figure 11: Sprint #2 Burndown Chart from Jira